

Q)A two member committee comprising of one male and one female member is to be constituted out of five males and three females. Amongst the females. Ms. A refuses to be a member of the committee in which Mr. B is taken as the member. In how many different ways can the committee be constituted?--> **14**

Q)In how many ways 2 students can be chosen from the class of 20 students?--> **190**

Q)Three gentlemen and three ladies are candidates for two vacancies. A voter has to vote for two candidates. In how many ways can one cast his vote?--> **15**

Q)A question paper has two parts, A and B, each containing 10 questions. If a student has to choose 8 from part A and 5 from part B, in how many ways can he choose the questions?--> **11340**

Q)In how many ways can 10 examination papers be arranged so that the best and the worst papers never come together?--> **$8 \cdot 9!$**

Q)In how many ways 4 boys and 3 girls can be seated in a row so that they are alternate?--> **144**

Q)In how many ways can 8 Indians, 4 American and 4 Englishmen be seated in a row so that all person of the same nationality sit together?--> **$3! 4! 8! 4!$**

Q)How many Permutations of the letters of the word APPLE are there?--> **60**

Q)A _____ of a set of distinct objects is an ordered arrangement of these objects.--> **permutation**

Q)Which among the following is not a permutation of the set $\{1,2,3\}$?--> **2, 1, 3**

Q)Which among the following is a 2-permutation of the set $\{1,2,3\}$?--> **2, 3**

Q)The number of 2-permutations possible for a set with 3 elements are--> **6**

Q)There are 10 people among whom two are brothers. The total number of ways in which these persons can be seated around a round table so that exactly one person sit between the brothers, is _____.--> **$2! \cdot 7!$**

Q)A letter lock consists of 4 rings, each ring contains 9 non-zero digits. This lock can be opened by setting four digit codes with the proper combination of each of the 4 rings. Maximum how many codes can be formed to open the lock?--> **9^4**

Q)In a party every person shakes hands with every other person. If there are 105 hands shakes, find the number of person in the party.--> **15**

Q)The number of positive integers which can be formed by using any number of digits from 0,1,2,3,4,5 without repetition.--> **1630**

Q)How many permutations of the letters ABCDEFGH contain the string ABC?--> **720**

Q)A _____ of elements of a set is an unordered selection of elements from the set.--> **combination**

Q)Which among the following is called r^{th} binomial coefficient?--> **nC_r**

Q)What is the coefficient of $x^{12}y^{13}$ in the expansion of $(x+y)^{25}$?--> **5200300**

Q)Suppose that there are eight runners in a race. The winner receives a gold medal, the second place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different chances are there to award these medals, if all possible outcomes of the race can occur and there are no ties?--> **336**

Q)Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?--> **5040**

Q)The number of r-permutations possible for a set with n distinct elements are--> **$n!/(n-r)!$**

Q)How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?--> **970200**

Q)For any positive integer n, the coefficient of x^n in the expansion of $(x+y)^n$ is always--> **N**

Q)A class contains 30 girls and 50 cricket players. How many students in the class are either girls or cricket players?--> **insufficient information**

Q) $|A \cup B|$ ---> **$|A| + |B| - |A \cap B|$**

Q)In a school every student can play either cricket or chess, or both. The number of students who can play cricket is 25; the number of students who can play chess is 13; and the number of students who can play both cricket and chess is 8. How many students are there in the school?--> **30**

Q)Pascals identity triangle is an application of--> **binomial theorem**

Q)What is the coefficient of x^{71} in $(1+x)^{11}$?--> **330**

Q)For a non-negative integer n, what is the sum of coefficients in the expansion of $(x+y)^n$?--> **2^n**

Q)For a non-negative integer n, what is the value of $\sum_{k=0}^n \binom{n}{k} 2^k$?--> **3^n**

Q)How many ways can 4 prizes be given away to 3 boys, if each boy is eligible for all the prizes?--> **24**

Q)If $5^n P_3 = 4^{n+1} P_3$, find n.--> **14**

Q)If ${}^6P_r = 360$ and ${}^6C_r = 15$, find r.--> **4**

Q)If there are _____ students in a class, then at least two of them will have names starting with same letter.--> **27**

Q) $|A \cup B \cup C|$ ---> **$|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$**

Q)A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both

Spanish and Russian, and 14 have taken courses in both French and Russian. If 2092 students have taken at least one of Spanish, French, and Russian, how many students have taken a course in all three languages?--> **7**

Q)How many positive integers not exceeding 1000 are divisible by 7 or 11?--> **220**

Q)Suppose that there are 1807 students at a school. Of these, 453 are taking a course in physics, 567 are taking a course in mathematics, and 299 are taking courses in both physics and mathematics. How many are not taking a course either in physics or in mathematics?--> **1508**

Q)How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 marks?--> **101**

Q)There are 366 people born in a non-leap year. At least how many of them will have birthday on same date?--> **2**

Q)A function f from a set with $k + 1$ or more elements to a set with k elements is not--> **one-to-one**

Q)The pigeonhole principle is also called--> **Dirichlet drawer principle**

Q)If there are 100 people, among them at least how many were born in same month?--> **9**

Q)If N objects are placed into k boxes, then there is at least one box containing at least $\frac{N}{k}$ objects.--> **$\frac{N}{k}$**

Q)How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?--> **9**

Q)What is the minimum number of students required in a class to be sure that at least six students will receive the same grade, if there are five possible grades, A, B, C, D, and F?--> **26**

Q)Given that $a_0 = 1$, $a_n = n + (-1)^n a_{n-1}$ for $n \geq 2$ What is the value of a_4 ?--> **5**

Q)_____ is an equation that recursively defines a sequence where the next term is a function of the previous terms.--> **recurrence relation**

Q)Which of the following recurrence relation gives solution for Towers of Hanoi problem?--> **$F_n = 2F_{n-1} + 1$**

Q)Which of the following recurrence relation find the n^{th} Fibonacci number?--> **$F_n = F_{n-1} + F_{n-2}$**

Q)A sequence that satisfies a recurrence relation is known as--> **solution**

Q)The number of bit strings of length 5 that do not have two consecutive 0s are--> **13**

Q)According to Pigeonhole Principle, if k is a positive integer and $k + 1$ or more objects are placed into k boxes, then there is at least one box contains _____ object(s).--> **2 or more**

Q)The Pigeonhole Principle deals with--> **more pigeons than pigeonholes**

Q)One of the roots for the recurrence relation $a_r = a_{r-1} + a_{r-2}$ is:--> **$[(1+\sqrt{5})/2]^r$**

Q)Which of the following is not a method for solving recurrence relations?--> **binomial theorem**

Q)The recurrence relation $T(n) = T(n-1) + c$, can be solved using--> **repeated substitution**

Q)Consider the series $A(X) = \sum_{r=0}^{\infty} a_r X^r$. Here $A(X)$ is known as--> **formal power series**

Q)One or more initial values required to solve recurrence relation are known as--> **boundary conditions**

Q)A characteristic equation of k^{th} degree has _____ characteristic roots.--> **k**

Q)_____ represents sequences where each term of a sequence is expressed as a coefficient of a variable x in a formal power series.--> **Generating Function**

Q)An equation relating a_r , to one or more of a_i , $i < r$ is called a--> **recurrence relation**

Q)Consider the series $A(X) = \sum_{r=0}^{\infty} a_r X^r$, $CA(X) = \sum_{r=0}^{\infty} C a_r X^r$

Q)Consider the series $A(X) = \sum_{r=0}^{\infty} a_r X^r$ and $B(X) = \sum_{r=0}^{\infty} b_r X^r$. $A(X) + B(X) = \sum_{r=0}^{\infty} (a_r + b_r) X^r$

Q)A formal power series is also known as a--> **ordinary generating function**

Q)The constant term in $A(X)$ is 2 and in $B(X)$ is 3. What will be the constant term in $A(X)B(X)$?--> **6**

Q)Consider the series $A(X) = \sum_{r=0}^{\infty} a_r X^r$. Here $a_0 X^0$ is known as--> **constant term**

Q)Consider the series $A(X) = \sum_{r=0}^{\infty} a_r X^r$ and $B(X) = \sum_{r=0}^{\infty} b_r X^r$. $A(X) = B(X)$ if and only if--> **$a_r = b_r \forall r \geq 0$**

Q)Consider the series $A(X) = \sum_{r=0}^{\infty} a_r X^r$. Here $a_i X^i$ is known as--> **term of degree i**

Q)Consider the series $A(X) = \sum_{r=0}^{\infty} a_r X^r$. Here a_i is known as--> **coefficient**

Q)Given $A(X) = 2 - 5X^3 + 7X^4 + 3X^8$ and $B(X) = 3 - 6X^4 + 8X^5 + 9X^8$. What is the coefficient of X^9 term in $A(X)B(X)$?--> **56**

Q)Consider the series $A(X) = \sum_{r=0}^{\infty} a_r X^r$, $B(X) = \sum_{r=0}^{\infty} b_r X^r$ and $C(X) = \sum_{r=0}^{\infty} c_r X^r$. What will be the coefficient of n^{th} degree term in $A(X)B(X)C(X)$?--> **$\sum_{i+j+k=n} a_i b_j c_k$**

Q)The generating function for a_r = the number of integral solutions to the equation $e_1 + e_2 + e_3 = r$ if $0 \leq e_i \leq 3$ for each i is _____ --> **$(1+X+X^2+X^3)^3$**

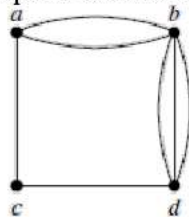
Q)If $A(X)$ is a formal power series and there exists another formal power series such that $A(X)B(X)=1$, the $B(X)$ is said to be _____ of $A(X)$.--> **multiplicative inverse**

Q)Given $A(X) = 2 - 5X^3 + 7X^4 + 3X^8$ and $B(X) = 3 - 6X^4 + 8X^5 + 9X^8$. What is the coefficient of X^4 term in $A(X)B(X)$?--> **9**

Q)Given $A(X) = 2 - 5X^3 + 7X^4 + 3X^8$ and $B(X) = 3 - 6X^4 + 8X^5 + 9X^8$. What is the coefficient of X^8 term in $A(X)B(X)$? --> **-67**

Q)Let P represent the principal borrowed from a bank and r be the interest rate per year. Which of the below recurrence relation finds the amount due after n years?--> **$a_n = a_{n-1} + r a_{n-1}$**

- Q) Consider the series $A(X) = \sum_{r=0}^{\infty} a_r X^r$ and $B(X) = \sum_{r=0}^{\infty} b_r X^r$. What will be the coefficient of n^{th} degree term in $A(X)B(X)$? --> $\sum_{r=0}^n a_r b_{n-r}$
- Q) Given $A(X) = 2 - 5X^3 + 7X^4 + 3X^8$ and $B(X) = 3 - 6X^4 + 8X^5 + 9X^8$. What is the coefficient of X^3 term in $A(X)B(X)$? --> **15**
- Q) Select proper solution for recurrence relation $a_n = a_{n-1} + n$ for $n \geq 1$. --> $a_n = c^n a_0 + \sum_{k=1}^n c^{n-k} k$
- Q) Select a solution for the recurrence relation $a_n = a_{n-1} + n$, where $a_0 = 2$. --> $a_n = \frac{n}{2}(n+1) + 2$
- Q) The generating function for a_r = the number of integral solutions to the equation $e_1 + e_2 + e_3 = r$ if $0 \leq e_i$ for each i is --> $(X + X^2 + 1)^3$
- Q) The generating function for a_r = the number of integral solutions to the equation $e_1 + e_2 + e_3 = r$ if $2 \leq e_i \leq 5$ for each i is --> $(X^2 + X^3 + X^4 + X^5)^3$
- Q) Recurrence relations are sometimes known as --> **difference equations**
- Q) If $f(n)$ is identically zero, then the recurrence relation is said to be --> **homogeneous**
- Q) Select a solution for the recurrence relation $a_n = a_{n-1} + 3^n$, where $a_0 = 1$. --> $a_n = \frac{3^{n+1} - 1}{2}$
- Q) The polynomial whose roots are in direct relationship to the form of solution for a recurrence relation is known as --> **characteristic polynomial**
- Q) Select a solution for the recurrence relation $a_n = a_{n-1} + 3n^2 + 3n + 1$, where $a_0 = 1$. --> $a_n = \frac{n^2(n+1)^2}{6} + 1$
- Q) Select a solution for the recurrence relation $a_n = a_{n-1} + n^3$, where $a_0 = 5$. --> $a_n = \frac{n^4}{4} + 5$
- Q) Select a solution for the recurrence relation $a_n = a_{n-1} + n^2$, where $a_0 = 7$. --> $a_n = \frac{n^3}{6} + 7$
- Q) Select a solution for the recurrence relation $a_n = a_{n-1} + 2n + 1$, where $a_0 = 1$. --> $a_n = \frac{(n+1)^2}{2} + 1$
- Q) Select a solution for the recurrence relation $a_n = a_{n-1} + 1/n(n+1)$, where $a_0 = 1$. --> $a_n = \frac{1}{n+1}$
- Q) Find the characteristic polynomial for the homogeneous recurrence relation whose general form is $a_n = B_1 2^n + B_2 n 2^n + B_3 3^n + B_4 n 3^n + B_5 6^n$. --> **$C(t) = (t-2)^2(t-3)^2(t-6)$**
- Q) Edges are used for connecting --> **vertices**
- Q) Find the characteristic polynomial for the homogeneous recurrence relation whose general form is $a_n = B_1 2^n + B_2 n 2^n + B_3 n^2 2^n$. --> **$C(t) = (t-2)^3$**
- Q) Find the characteristic polynomial for the homogeneous recurrence relation whose general form is $a_n = B_1 + nB_2 - n^2B_3$. --> **$C(t) = (t-1)^2$**
- Q) Find the characteristic polynomial for the homogeneous recurrence relation whose general form is $a_n = B_1 + nB_2 + n^2B_3$. --> **$C(t) = (t-1)^3$**
- Q) Find the characteristic polynomial for the homogeneous recurrence relation whose general form is $a_n = B_1 2^n + B_2 3^n$. --> **$C(t) = (t-2)(t-3)$**
- Q) Find the characteristic polynomial for the homogeneous recurrence relation whose general form is $a_n = B_1 2^n + B_2 n 2^n$. --> **$C(t) = (t-2)^2$**
- Q) A path with same source and destination is known as a --> **cycle**
- Q) A graph with all directed edges is known as --> **digraph**
- Q) A loop in a graph is --> **an edge that connects a vertex to itself**
- Q) Which of the following can be empty is a graph? --> **no. of vertices**
- Q) An infinite graph contains infinite number of --> **edges**
- Q) If a graph contains multiple edges connecting same vertices, then the graph can be called as --> **multi graph**
- Q) For representing which among the following, graph is not a suitable structure? --> **relationship between presidents of India from 1950 to 2017**
- Q) Contribution of loop at vertex in the degree of that vertex is --> **2**
- Q) In an undirected graph with x vertices and y edges, the sum of degrees of all vertices is --> **$2y$**
- Q) The degree of a vertex in an undirected graph is --> **number of edges incident to the vertex**
- Q) What type of edges is suitable for representing one-way roads? --> **directed edges**
- Q) A graph which allows both multiple edges and loops is known as --> **pseudo graph**



- Q) Choose the right type for the graph given below: --> **multi graph**
- Q) Two vertices u and v of a graph are said to be adjacent iff u and v are connected by a --> **edge**
- Q) A path is a sequence of --> **edges**

- Q) Two vertices in a graph are said to be connected if there exists at least one _____ between them.--> **path**
- Q) How many elements are required in an adjacency matrix, to represent a graph with n vertices and m edges?--> **n^2**
- Q) In an undirected graph, number of vertices with odd degree are always--> **even**
- Q) In a directed graph with n vertices and m edges, the sum of out degrees of all vertices is--> **m**
- Q) In a complete (undirected) graph with n vertices, number of edges is equal to _____--> **$n(n-1)/2$**
- Q) In a simple graph, if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 , then the graph is said to be--> **bipartite graph**
- Q) A graph without cut vertices or cut edges is known as a _____ graph.--> **biconnected**
- Q) Isomorphism of two simple graphs is a _____ relation.--> **equivalence**
- Q) A vertex whose removal produces a graph with more connected components than in the original graph is called--> **articulation point**
- Q) A circuit/cycle is a path whose end points are--> **same vertex**
- Q) A path is said to be a simple path if it doesn't have same _____ more than once.--> **edge**
- Q) A graph which is not connected is known as _____ graph--> **disjoint**
- Q) An edge whose removal produces a graph with more connected components than in the original graph is called--> **bridge**
- Q) An Euler circuit in a graph G is a simple circuit containing every _____ of G .--> **edge**
- Q) _____ path in a graph G is a simple path containing every edge of G .--> **Euler**
- Q) A directed graph is _____ if there is a path from a to b and from b to a for every pair of vertices a & b in the graph.--> **strongly connected**
- Q) Incidence matrix shows relationship between--> **edges and vertices**
- Q) Which among the following representations of graphs, wastes storage space in case of sparse graphs (graphs with less number of edges)?--> **adjacency matrix**
- Q) If two graphs have exactly the same form, the two graphs are said to be--> **isomorphic**
- Q) How many non-isomorphic connected simple graphs are there with 4 vertices?--> **8**
- Q) Which among the following algorithm identifies the shortest path between a pair of vertices in a graph?--> **Dijkstra's algorithm**
- Q) A graph is called _____ if it can be drawn in the plane without any edges crossing.--> **planar**
- Q) A graph with each edge associated with a numerical cost is known as a _____--> **weighted graph**
- Q) A connected multigraph with at least two vertices has an Euler circuit if and only if each of its vertices has _____.--> **even degree**
- Q) A simple path in a graph G that passes through _____ exactly once is called a Hamilton path.--> **every vertex**
- Q) A simple circuit in a graph G that passes through every vertex exactly once is called a _____ circuit.--> **Hamiltonian**
- Q) In a simple undirected graph, Hamiltonian Cycle exists only if the degree of each vertex is at least--> **2**
- Q) The least number of colors needed for coloring of a graph is known as--> **chromatic number**
- Q) The chromatic number of any planar graph is not greater than--> **4**
- Q) In graph coloring, colors are assigned such that no two _____ have same color.--> **adjacent vertices**
- Q) Planar graphs are useful in designing--> **electronic circuits**
- Q) Let G be a connected planar simple graph with e edges and v vertices. Let r be the number of regions in a planar representation of G . Then r =--> **$e-v+2$**
- Q) Suppose that a connected planar simple graph has 20 vertices, each of degree 3. Into how many regions does a representation of this planar graph split the plane?--> **12**
- Q) If G is a connected planar simple graph, then G has a vertex of degree not exceeding--> **5**

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